

Effect of lactate and β -hydroxybutyrate infusions on brain metabolism in the fetal sheep

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Abstract

Brain uptake of substrates other than glucose has been demonstrated in neonatal but not fetal animals *in vivo*. This study was undertaken to investigate the ability of the fetal sheep brain to use potential alternative substrates when they were provided in increased amounts. Brain substrate uptake was measured in chronically catheterised fetal sheep during 2-h infusions of neutralised lactate ($n=12$) or β -hydroxybutyrate ($n=12$). Despite large increases in fetal arterial lactate and β -hydroxybutyrate during the respective infusions, no significant uptake of either substrate was demonstrated.

However during both types of infusion, the brain arterio-venous difference for glucose decreased 30% ($P<0.05$). Since the brain arterio-venous difference for oxygen was unchanged, and blood flow to the cerebral hemispheres (measured in 11 studies) was also unchanged, the infusions appeared to cause a true decrease in brain glucose uptake. This decrease paralleled the rise in lactate concentration during lactate infusions, and the rise in lactate and butyrate concentrations during the butyrate infusions. Both substrates have metabolic actions that may inhibit brain glucose uptake. We speculate that the deleterious effects of high lactate and ketone states in the perinatal period may in part be due to inhibition of brain glucose uptake.

1985), and in children with glucose-6-phosphatase deficiency (Fernandes, Berger & Smit, 1984).

Brain uptake of these alternative fuels is usually proportional to their circulating concentrations (Kraus, Schlenker & Schwedesky, 1974; Persson, Settergren & Dahlquist, 1972; Hellman, Vannucci & Nardis, 1982). Furthermore, at any given arterial concentration, brain uptake of ketones is greater in the neonate than in the older animal (Hawkins, Williamson & Krebs, 1971; Kraus, *et al.*, 1974; Dahlquist & Persson, 1976). This suggests that the brain may be peculiarly adapted to the use of alternative fuels in the neonatal period, when glucose supplies are most likely to be in jeopardy, and circulating levels of alternative fuels may be high.

It is not known if this capacity to use alternative fuels develops before birth *in vivo*. Glucose uptake has been shown to account for all fetal sheep brain oxidative fuel requirements (Tsoulos, Schneider, Colwill, Meschia, Makowski & Battaglia, 1972; Makowski, Schneider, Tsoulos, Colwill, Battaglia & Meschia, 1972) and no significant uptake of lactate or ketones has been demonstrated in the fetal sheep brain *in vivo* (Jones, Burd, Makowski, Meschia & Battaglia, 1975). However these studies have not been performed under conditions where the circulating levels of potential alternative substrates have been elevated; the only situation in which such uptake has been demonstrated postnatally. Furthermore, there have been no studies investigating the possible changes in alternative fuel use by the fetal brain during gestation.

We therefore undertook studies in the chronically-catheterised fetal sheep to determine the ability of the fetal brain to use potential alternative substrates (lactate and β -hydroxybutyrate) when they were provided in increased amounts. We also investigated whether this capacity to use alternative fuels changed with gestation during the last trimester of pregnancy.

Experimental

Animal preparation

Surgery was performed on 14 mixed western breed pregnant ewes at 109–133 days gestation (term=145

Introduction

Glucose has long been recognised as the primary metabolic fuel for the brain under normal circumstances. However, under special circumstances, alternative fuels may be used. The ketones, β -hydroxybutyrate and acetoacetate, contribute up to 60% of human brain oxidative fuel requirements during prolonged starvation (Owen, Morgan, Kemp, Sullivan, Herrera & Cahill, 1967). Lactate may be a significant brain fuel in the newborn baboon (Levitsky, Fisher, Paton & Delannoy, 1977) and rat (Medina,

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days). All but one carried single fetuses. Spinal anaesthesia was induced with 1% tetracaine and analgesia and sedation maintained with intravenous pentobarbitone and ketamine as required. Polyvinyl catheters were placed in the maternal femoral artery and vein. Hysterotomy was performed via a mid-line abdominal incision and fetal catheters placed under local anaesthesia using 1% xylocaine. Polyvinyl catheters were placed in the fetal aorta and inferior vena cava via the tarsal artery and vein. An open-ended amniotic catheter was placed and the uterine incision closed. Through a separate uterine incision, catheters were placed in a fetal brachial artery and vein at the mid-humeral level, and advanced so the tips lay close to the brachiocephalic trunk and superior vena cava. The crown of the fetal head was then delivered to the uterine incision and the sagittal sinus catheterised using a 18 gauge teflon catheter connected to a polyvinyl extension (Iwamoto, Rudolph & Bristow, 1984, Tsoulos *et al.*, 1972). The catheter junction and skull defect were sealed with dental cement and the scalp closed.

Following surgery animals were housed in individual pens with free access to water and alfalfa concentrates. They were brought to the laboratory daily in study carts, where they were also free to stand or sit and had free access to food and water. Catheters were flushed daily with heparin (1,000 units/ml). Antibiotics (1 Mega-unit penicillin and 100 mg gentamicin) were given intravenously to the ewe and into the amniotic fluid daily post-operatively for five days, and then once at the end of each study.

At the end of all studies the ewe was killed with an overdose of thiopentone. The position of all catheters was checked, and the fetus dissected. The brain was removed, separated into cerebral hemispheres, cerebellum and brain stem, weighed, and placed in formalin.

Study protocols

All studies were performed at least five days after surgery on animals who were in good condition, feeding well, with normal blood gas and metabolite concentrations. Studies were performed as close as possible to 120, 130 and 140 days gestation. Each study consisted of a 2-h fetal infusion of L-lactic acid, sodium β -hydroxybutyrate or normal saline, followed by a 2-h recovery period. L-lactic acid and sodium β -hydroxybutyrate were made up in distilled water, neutralised to pH 7.4, and sterilised by passage through a 0.45 μ filter (Falcon, Oxnard, California). Infusate concentration was calculated to give 0.1 mmol/min per kg of estimated fetal weight when infused at a rate of 0.1 ml/min using a syringe pump (Harvard, So. Natick, MA). For control infusions, sterile isotonic saline was infused at the same rate. Infusions were given via the brachial vein catheter, beginning with a 1-ml bolus.

For lactate and β -hydroxybutyrate studies blood

samples were taken from the brachial artery, sagittal sinus and maternal arterial catheters for blood gas and metabolite assays immediately before, and at hourly intervals during the 4-h study period. Radio-labelled microspheres (15 μ , Minnesota Mining & Manufacturing, St. Paul, Minnesota) were injected via the femoral venous catheter before and at the end of the two hour infusion period for measurement of brain blood flow, with reference samples drawn from the brachial artery. For control studies, blood samples of the same total volume were taken on the same time schedule from the fetal and maternal femoral arterial catheters, to be sure that metabolic changes during the substrate infusions were due to the infusate and not the study protocol. At the end of each study fetal blood loss was replaced with an equivalent volume of maternal blood. At least two days were allowed between studies on the same animal.

Animals were alternately assigned to lactate or β -hydroxybutyrate infusions initially, and not more than one infusion of each type was done on each animal at each gestational age. A total of 30 studies were performed on 14 fetuses; 12 lactate infusions (6 at 120 days, 5 at 130 days and 1 at 140 days), 12 β -hydroxybutyrate infusions (5 at 120 days, 5 at 130 days and 2 at 140 days), and 6 control infusions (1 at 120 days, 3 at 130 days and 2 at 140 days). The average age at study was 126 ± 6 days (mean $\pm$ SD). All fetuses were normally grown, as judged by weight and length measurements at post mortem (Barcroft 1946).

Samples

Blood gas samples (0.4 ml) were drawn into 1 ml heparinised glass syringes on ice, and analysed immediately in duplicate using a Corning 158 blood gas analyser (Corning Diagnostic, Medfield, MA) and a Radiometer OSM 2 haemoxymeter (Radiometer, Copenhagen). Blood samples for metabolite assays (1.4 ml) were drawn into heparinised syringes and immediately deproteinised on ice using perchloric acid. Samples were centrifuged at 4°C, and the supernatant decanted and frozen at -70°C until assay.

All assays were performed in duplicate, and all samples from one experiment assayed in the same batch. Glucose was measured using a glucose oxidase method and lactate and pyruvate were measured using standard enzymatic spectrophotometric techniques (Sigma Chemicals, St. Louis, MO). β -hydroxybutyrate assay was based on the spectrophotometric method reported by Kientsch-Engel & Siess (1983). The mean percentage errors between duplicate samples were: glucose 3.1; lactate 2.0; pyruvate 8.7; β -hydroxybutyrate 4.3.

For measurement of brain blood flow, tissues were carbonised at 325°C for at least 24 hours, powdered and packed into vials for counting on an Ultima II Multichannel Pulse Height Analyser (Ino Tech. Co.,

Fort Atkinson, Wisconsin). Up to 9 different isotopes were used for each animal: ^{51}Cr , ^{85}Sr , ^{95}Nb , ^{54}Mn , ^{113}Sn , ^{65}Zn , ^{153}Gd , ^{114}In and ^{57}Co . Time elapsed between the first microsphere injection and collection of tissues was not more than 20 days, and no significant loss of microspheres from tissue beds *in vivo* has been demonstrated over this period (Consigny, Verrier, Payne, Edelist, Jester, Baer, Vlahakes & Hoffman, 1982; Charlton, unpublished observations). Blood flow to the brain was calculated from counts in the reference sample using methods developed and validated in this laboratory (Heymann, Payne, Hoffman & Rudolph, 1977; Rudolph & Heymann, 1980).

Calculations

Blood oxygen contents were calculated from measured PO_2 , haemoglobin and oxygen saturations (McKinney & Harris, 1978). Metabolite concentrations were all expressed as the mean of duplicate samples; if duplicates were widely discrepant, both values were discarded. Brain substrate uptakes were calculated as the substrate arterio-venous difference across the brain $\times$ blood flow to the cerebral hemispheres.

Preinfusion metabolite and arterio-venous differences were compared to values obtained during and after infusions using the Wilcoxon Signed Rank test for paired data. Mean metabolite concentrations and arterio-venous differences were compared at different gestational ages using the Mann-Whitney U test. Arterio-venous differences were tested for significant difference from zero using 95% confidence intervals (Zar, 1984). All values are expressed as the mean $\pm$ SD.

Results

Lactate infusions

During the 12 fetal lactate infusions fetal blood lactate concentrations increased 4-fold (1.64 ± 0.50 to 6.80 ± 2.31 mM, Fig. 1) with no change in fetal pH, blood gases or oxygen content. The pre-infusion arterio-venous difference for lactate across the brain was not significantly different from 0, indicating no net brain uptake or release of lactate (Table 1). During the infusion the arterio-venous difference for lactate remained not different from 0, again indicating no significant net brain uptake of lactate.

Changes in fetal arterial pyruvate concentrations paralleled those of lactate (0.07 ± 0.03 at 0h to 0.18 ± 0.06 mM at 2 h, Fig. 1). There was a small brain release of pyruvate which increased during the first hour of the lactate infusion, and again during the first hour of the recovery period (Fig. 1).

Fetal arterial glucose concentration did not change during the study (1.44 ± 0.30 mM at 0h and 1.49 ± 0.26 mM at 2 h). However, the arterio-venous difference for glucose fell by 30% by the end of the lactate infusion (0.32 ± 0.13 mM to 0.20 ± 0.06 mM, $P < 0.01$), and recovered in the next 2 h (Fig. 1). In

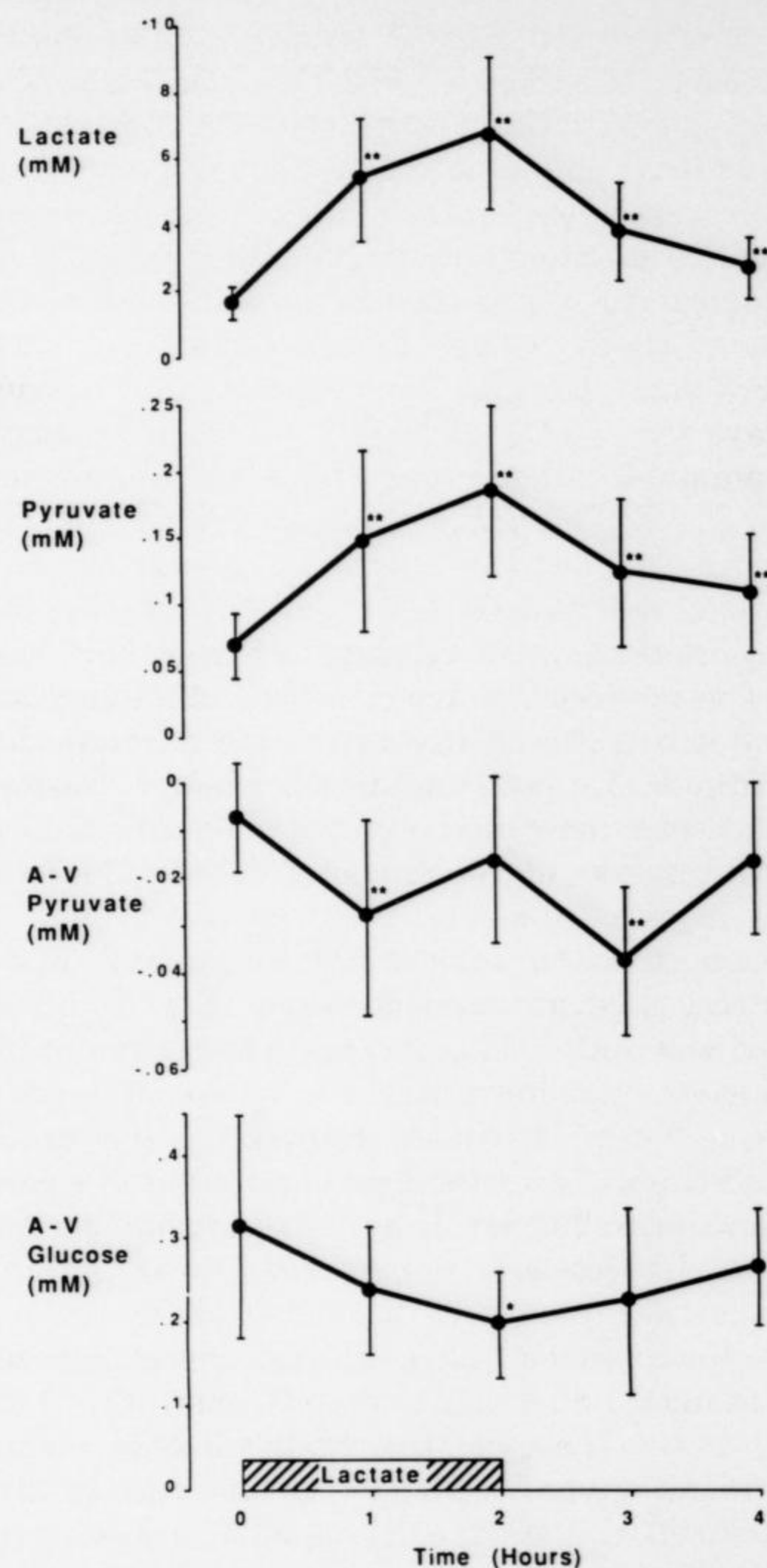


Fig. 1. Effects of lactate infusion on fetal arterial lactate and pyruvate concentrations, and on the arterio-venous differences for pyruvate and glucose across the fetal brain. * $P < 0.05$, ** $P < 0.01$ for comparison with value before beginning infusion (0 h). Period of lactate infusion is shown by hatched bar. Values are mean $\pm$ SD for 12 studies.

contrast, the arterio-venous difference for oxygen did not change (1.17 ± 0.40 mM at 0 h and 1.01 ± 0.13 mM at 2 h).

When studies done at 120 days ($n=6$) and 130 days ($n=5$) were compared, there were no gestational differences in the metabolic responses to lactate infusion. Although a higher arterial lactate concentration was reached after 2 h at 130 days than at 120 days (8.30 ± 2.05 vs 5.09 ± 1.05 mM, $P < 0.05$), in neither group was brain lactate uptake seen and in both, brain arterio-venous differences for glucose fell (Table 2). The single study at 140 days was not different from those at 120 days and 130 days.

β -hydroxybutyrate infusions

During the 12 fetal β -hydroxybutyrate infusions the mean fetal butyrate concentrations increased from

Table 1. Initial fetal arterial substrate concentrations and brain arterio-venous differences before lactate and β -hydroxybutyrate infusions.

	Arterial concentrations (mM)	Arterio-venous Difference (mM)
<i>Lactate Infusions</i>		
Glucose	1.44 ± 0.30	0.32 ± 0.13
Oxygen	2.92 ± 0.60	1.17 ± 0.40
Lactate	1.64 ± 0.50	0.07 ± 0.17
Pyruvate	0.07 ± 0.03	-0.01 ± 0.01
<i>β-hydroxybutyrate Infusions</i>		
Glucose	1.35 ± 0.26	0.29 ± 0.09
Oxygen	2.80 ± 0.74	1.05 ± 0.39
Lactate	1.48 ± 0.29	-0.05 ± 0.13
Pyruvate	0.06 ± 0.02	-0.01 ± 0.01
β -hydroxybutyrate	0.088 ± 0.024	-0.001 ± 0.006

Values are mean $\pm$ SD for 12 studies in each group.

0.09 ± 0.02 mM to 5.69 ± 1.62 mM at 2 h (Fig. 2). Once again, there was no change in fetal pH or blood gases. The pre-infusion arterio-venous difference for butyrate across the brain was not significantly different from 0 (Table 1) and did not change. Despite the increase in arterial concentrations there was no significant uptake of β -hydroxybutyrate by the fetal brain.

Butyrate infusion caused a rise in fetal blood lactate and pyruvate concentrations (Fig. 2) which persisted and continued to increase after the infusion had finished, but there was no uptake of lactate across the brain. Pyruvate release by the brain increased slightly over the first hour of the infusion, and again during the first hour of the recovery period, but these changes were not statistically significant (Fig. 2).

Once again, arterial blood glucose concentrations did not change (1.35 ± 0.26 mM at 0 h and 1.39 ± 0.32 mM at 2 h). However the brain arterio-venous difference for glucose fell 28% by the end of the infusion (0.29 ± 0.09 to 0.21 ± 0.06 mM at 2 h, $P < 0.05$). The brain arterio-venous difference for oxygen did not change (1.05 ± 0.39 mM at 0 h and 0.92 ± 0.23 mM at 2 h).

When studies done at 120 days ($n=5$) and 130 days ($n=5$) were compared, there were no gestational differences in the metabolic responses to butyrate infusion, although again a higher arterial lactate concentration was reached after 2 h at 130 days than at 120 days (3.73 ± 0.64 vs 2.75 ± 0.42 mM, $P < 0.05$, Table 2). The 2 studies at 140 days were similar to those at 120 days and 130 days.

Control infusions

During the 6 control saline infusions there were no changes in any of the fetal blood gas or metabolite concentrations measured. This indicates that the observed changes in pyruvate concentrations during lactate infusions, and in pyruvate and lactate concentrations during β -hydroxybutyrate infusions were not due to fetal blood sampling or other aspects of the experimental procedures.

Blood flow measurements and brain substrate uptake

Because of technical problems, blood flow data was incomplete. During the lactate studies mean blood

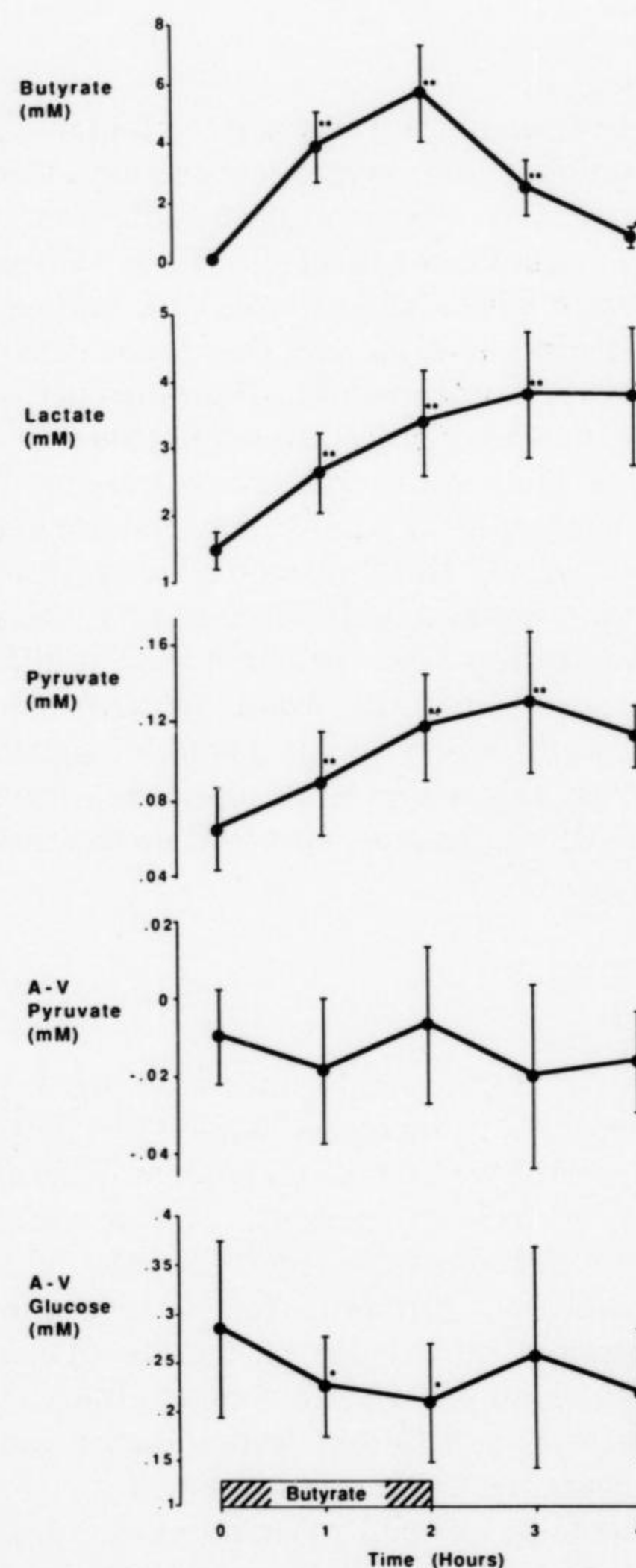


Fig. 2. Effects of β -hydroxybutyrate infusion on fetal arterial butyrate, lactate and pyruvate concentrations, and on the arterio-venous differences for pyruvate and glucose across the fetal brain. * $P < 0.05$, ** $P < 0.01$ for comparison with value before beginning infusion (0 h). Period of β -hydroxybutyrate infusion is shown by hatched bar. Values are mean $\pm$ SD for 12 studies.

flow to the cerebral hemispheres was 37 ± 12 ml/min ($n=8$) before and 40 ± 11 ml/min ($n=8$) after the 2-h lactate infusion. For the 7 studies in which both values were obtained, there was no consistent change

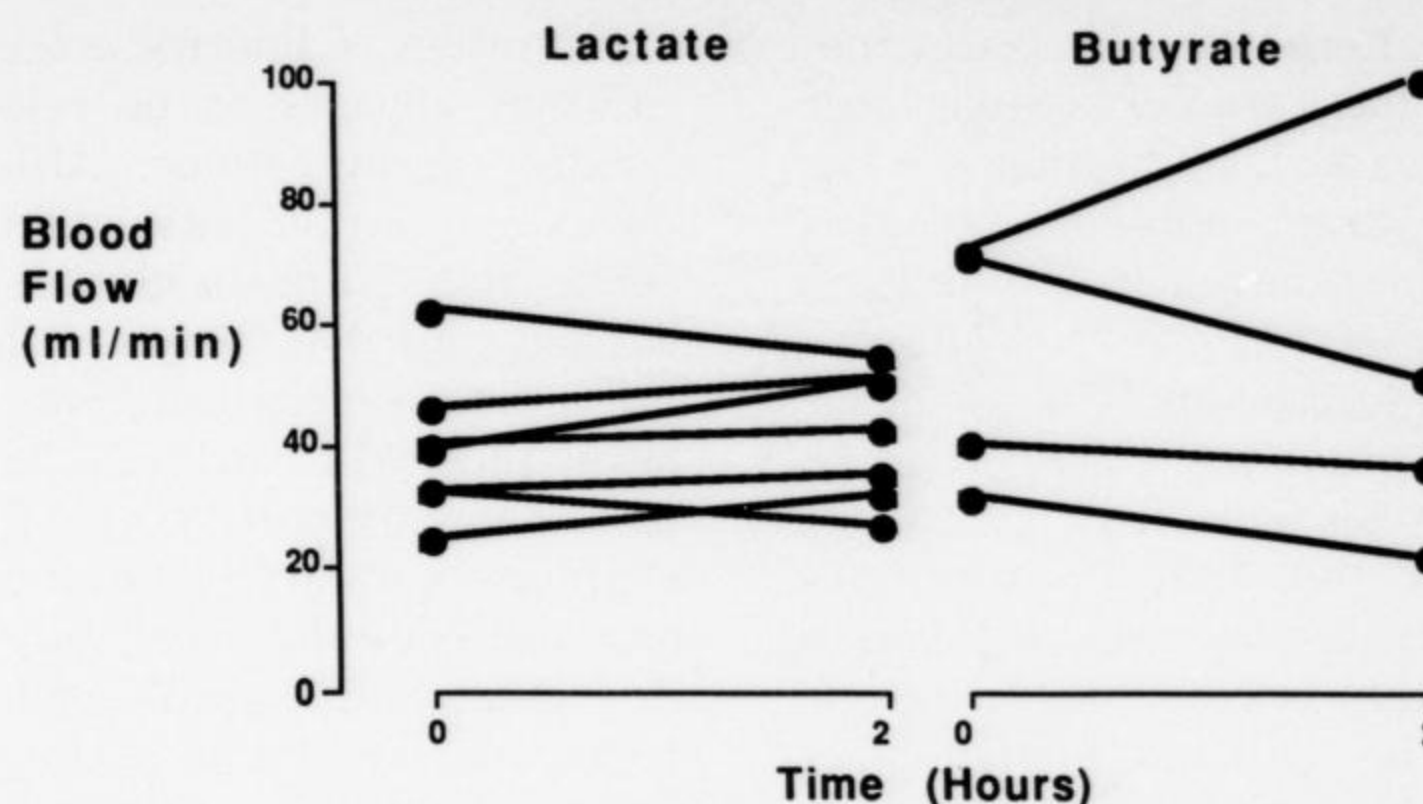


Fig. 3. Effects of lactate and β -hydroxybutyrate infusions on blood flow to the cerebral hemispheres. Changes in blood flow are shown for the 7 lactate and 4 butyrate studies in which measurements were obtained both before (0 h) and at the end of (2 h) the infusions.

in blood flow (Fig. 3). In these 7 studies brain glucose uptake fell 26% during the lactate infusion (12.1 ± 6.1 at 0 h to 8.9 ± 3.1 $\mu\text{mol/min}$ at 2 h, $P=0.062$) while oxygen uptake was unchanged (43.9 ± 14.7 at 0 h and 39.9 ± 13.5 $\mu\text{mol/min}$ at 2 h).

During the β -hydroxybutyrate studies mean blood flow to the cerebral hemispheres was 46 ± 22 ml/min ($n=7$) before and 50 ± 30 ml/min ($n=5$) after the infusion. For the 4 studies in which both values were obtained, there was again no consistent change in blood flow (Fig. 3). In these 4 studies the 24% fall in mean brain glucose uptake was not statistically significant (15.5 ± 7.4 at 0 h to 11.8 ± 8.5 $\mu\text{mol/min}$ at 2 h, $P=0.265$). Oxygen uptake also showed no significant change (58.7 ± 19.8 at 0 h to 48.0 ± 35.8 $\mu\text{mol/min}$ at 2 h).

Maternal measurements

Maternal blood gas and metabolite concentrations at the beginning of the studies were: pH 7.47 ± 0.03 ; PO_2 105.6 ± 8.4 mmHg; glucose 3.43 ± 0.39 mM; lactate 0.90 ± 0.29 mM; pyruvate 0.05 ± 0.02 mM; β -hydroxybutyrate 0.77 ± 0.19 mM.

There were no changes in maternal blood gas or metabolite concentrations during any of the studies, except for a small fall in the maternal lactate concentration over the first hour of the infusions (0.90 ± 0.29 mM to 0.72 ± 0.13 mM, $P<0.01$). This level then remained constant for the remainder of the studies.

Discussion

These studies aimed to determine the ability of the fetal brain to use potential alternative substrates, lactate and β -hydroxybutyrate, when they were provided in increased amounts. During 2-h infusions we were unable to demonstrate significant uptake of either substrate by the third trimester fetal sheep brain *in vivo*. Although these studies do not exclude the possibility of brain uptake of small amounts of alternative substrates, uptake of quantities sufficient to make a significant contribution to brain oxidative

fuel requirements should have been detected using this methodology.

In view of the ability of newborns in many species to use alternative substrates for brain metabolism, this finding is perhaps surprising. There are a number of possible reasons why such uptake could not be demonstrated in these studies. Concentrations of alternative substrates achieved during the infusions may have been too low, since uptake has been shown to be proportional to the circulating concentrations (Kraus *et al.*, 1974; Hawkins *et al.*, 1971; Hellman *et al.*, 1982). However the concentrations reached at the end of the infusions were higher than those associated with significant brain uptake in post-natal animals (Settergren, Lindblad & Persson, 1976), Levitsky *et al.*, 1977). Furthermore, neonatal brain uptake of alternative substrates is reportedly greater than that of the adult brain at any given concentration (Kraus *et al.*, 1974; Dalquist & Persson, 1976; Hawkins *et al.*, 1971). The arterial concentrations reached were also similar to the estimated K_m of the transport mechanism for brain ketone uptake in the adult rat (Conn, Fell & Steele, 1983), and three times higher than that estimated for lactate (Pardridge, Connor & Crawford, 1975).

Another possible reason for our failure to demonstrate uptake in these studies could be that the alternative substrates were provided for too short a time. In rats, the brain ketone uptake mechanism is inducible over several days of starvation (Zivin & Snarr, 1972; Gjedde & Crone, 1975), and may be regulated by prolonged changes in circulating concentrations of ketones (Sarna, Bardbury, Cremer, Lai & Teal, 1979; Moore, Lione, Sugden & Regen 1976). The enzymes involved in ketone metabolism also increase during persistent ketosis, both postnatally (Land, Booth, Berger & Clark, 1977; Page, Krebs & Williamson, 1971) and prenatally (Thaler, 1972). Thus prolonged exposure to alternative substrates may be necessary before significant brain uptake can be detected.

Fetal brain uptake of alternative substrates may also have not occurred in these studies because brain glucose supply was adequate. Lactate makes a significant contribution to brain substrate requirements in newborn dogs only during insulin-induced hypoglycaemia (Hellman *et al.*, 1982). Additional studies in fetuses whose glucose supply is restricted, such as those suffering intrauterine growth retardation, would help to explore this possibility.

Finally, the ability of the fetal brain to use alternative substrates may develop only very late in gestation. The activity of enzymes required for ketone metabolism are low in the 32 week human fetus (Page & Williamson, 1971) and increases towards term in the fetal rat (Thaler, 1972). We were unable to demonstrate differences in fetal brain metabolism at the gestations studied, but too few studies were performed at term to exclude the possibility that this ability may develop in the last week of gestation. Further studies are required to clarify this question.

An unexpected finding in these studies was the 30% decrease in arterio-venous difference for glucose across the brain after two hour infusions of both lactate and β -hydroxybutyrate (Fig. 1 & 2, Table 2). This appears to reflect a true decrease in brain glucose uptake rather than an increase in blood flow, since there was no consistent change in brain blood flow during the infusions (Fig. 3), and there was a decrease in mean brain glucose uptake where it was measured. Furthermore, there was no evidence of the concomitant fall in brain arterio-venous difference for oxygen which would be expected if blood flow increased.

Inhibition of brain glucose uptake during lactate infusion appears to be related to the circulating lactate concentrations. Although there is a poor direct correlation between the arterial lactate concentration and brain glucose uptake ($r^2=0.195$), our data suggest that there is a threshold effect, whereby the arterio-venous difference for glucose across the brain falls when arterial lactate concentration exceeds approximately 3.5–4.5 mM (Fig. 4). Inhibition of glucose uptake by similar concentrations of lactate has also been described *in vitro* in rat brain slices (Ide, Steinke & Cahill, 1969; Fernandez & Medina, 1986), and *in vivo* in young rats (Miller, Kinney & Staton, 1984).

The mechanism for inhibition of brain glucose uptake by lactate is not known. Lactate and glucose are transported across the blood brain barrier by different carriers and inhibition is therefore unlikely to be because of competition for entry into the brain (Pardridge *et al.*, 1975). However, if lactate is entering the fetal brain, albeit in amounts too small to be detected in these studies, metabolism of lactate may be inhibiting glycolysis. Conversion of lactate to pyruvate generates NADH, which in increased concentrations inhibits pyruvate metabolism to acetyl CoA and entry into the citric acid cycle (Lehninger, 1982). Pyruvate accumulation is suggested by the significant increase in efflux of pyruvate from the brain during the first hour of lactate infusion (Fig. 1). There is then a second rise in pyruvate efflux from the brain during the first hour after the infusion stops, as blood lactate and pyruvate levels fall and brain glucose uptake rises again. A similar efflux of

Table 2. Fetal arterial substrate concentrations and brain arterio-venous differences before (0 h) and after (2 h) lactate and β -hydroxybutyrate infusions at 120 and 130 days gestation.

		Arterial concentrations (mM)		Arterio-venous difference (mM)	
		0 h	2 h	0 h	2 h
<i>Lactate infusions</i>					
Lactate	120 days	$1.34 \pm 0.26^\dagger$	$5.09 \pm 1.05^{*\dagger}$	0.01 ± 0.02	0.03 ± 0.09
	130 days	1.77 ± 0.35	$8.30 \pm 2.05^*$	0.04 ± 0.04	0.06 ± 0.08
Glucose	120 days	1.58 ± 0.20	1.60 ± 0.16	0.29 ± 0.18	0.19 ± 0.08
	130 days	1.28 ± 0.36	1.36 ± 0.34	0.31 ± 0.04	$0.21 \pm 0.06^*$
Oxygen	120 days	3.10 ± 0.35	3.07 ± 0.46	1.06 ± 0.17	1.02 ± 0.13
	130 days	2.99 ± 0.54	2.69 ± 0.19	1.40 ± 0.51	$1.00 \pm 1.16^*$
Pyruvate	120 days	0.05 ± 0.02	$0.15 \pm 0.04^*$	-0.01 ± 0.01	-0.01 ± 0.02
	130 days	0.08 ± 0.02	$0.22 \pm 0.07^*$	-0.01 ± 0.01	-0.02 ± 0.01
<i>β-Hydroxybutyrate infusions</i>					
β -Hydroxybutyrate	120 days	0.10 ± 0.02	$4.82 \pm 1.06^*$	-0.002 ± 0.009	0.204 ± 0.244
	130 days	0.09 ± 0.03	6.30 ± 2.07	-0.001 ± 0.005	-0.132 ± 0.378
Lactate	120 days	1.33 ± 0.24	$2.75 \pm 0.42^{*\dagger}$	-0.09 ± 0.19	-0.07 ± 0.10
	130 days	1.55 ± 0.26	$3.73 \pm 0.64^*$	-0.03 ± 0.06	0.08 ± 0.18
Glucose	120 days	1.45 ± 0.35	1.51 ± 0.40	0.29 ± 0.07	0.19 ± 0.09
	130 days	1.26 ± 0.20	1.25 ± 0.28	0.24 ± 0.09	0.22 ± 0.03
Oxygen	120 days	3.07 ± 0.48	2.74 ± 0.39	1.09 ± 0.33	0.94 ± 0.10
	130 days	2.95 ± 0.77	2.88 ± 0.76	1.12 ± 0.52	1.03 ± 0.30
Pyruvate	120 days	0.07 ± 0.02	$0.11 \pm 0.02^*$	-0.01 ± 0.02	-0.02 ± 0.01
	130 days	0.07 ± 0.03	$0.13 \pm 0.04^*$	-0.01 ± 0.01	0.01 ± 0.02

* $P < 0.05$ for comparison with value before beginning infusion (0 h); $^\dagger P < 0.05$ for comparison with value at 130 days gestation. Values are Mean $\pm$ S.D. for: lactate infusions 120d – 6 studies at 121 ± 0.8 days gestation, 130d – 5 studies at 129 ± 1.1 days gestation. For butyrate infusions 120d – 5 studies at 119 ± 2.2 days gestation, 130d – 5 studies at 129 ± 1.1 days gestation.

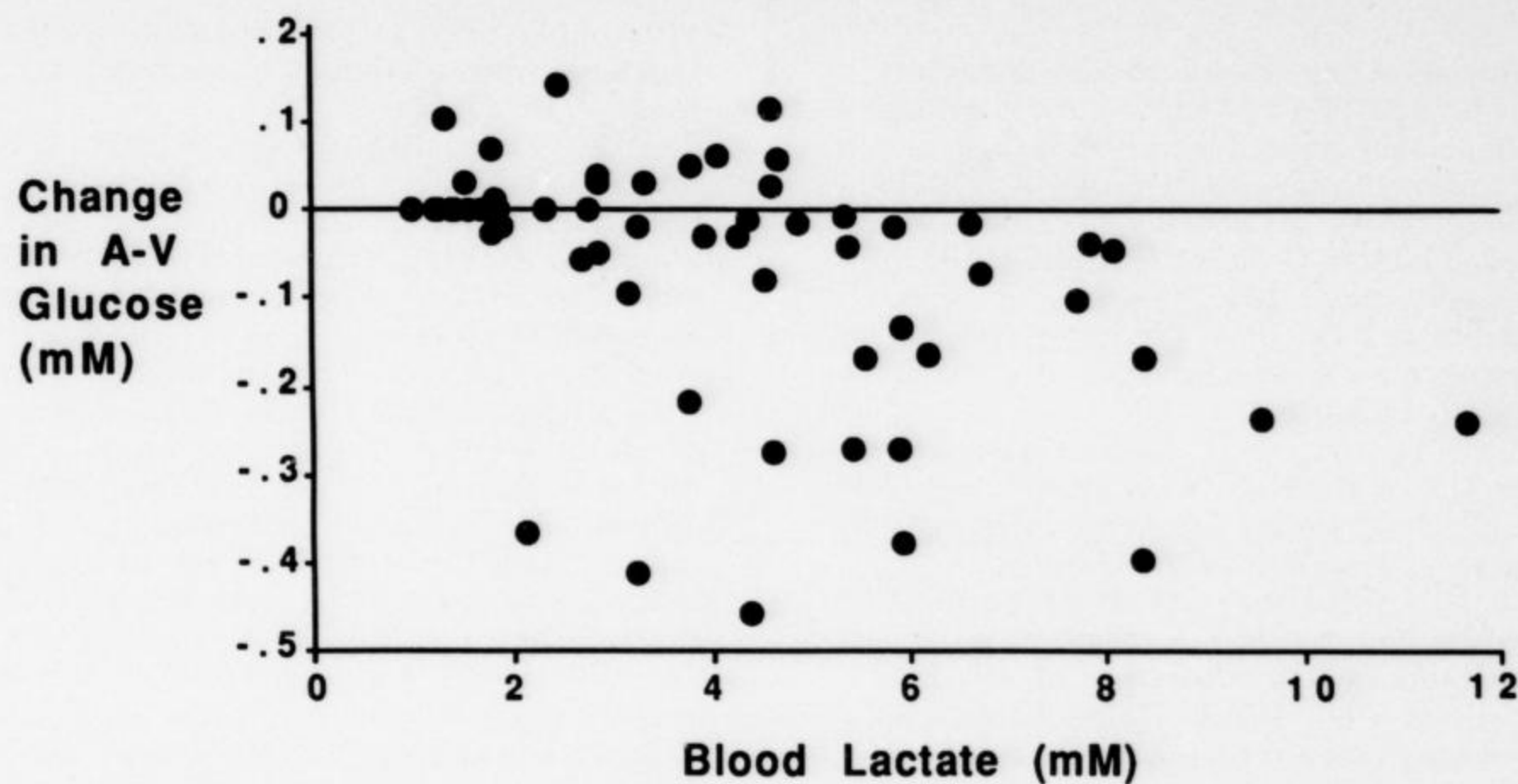


Fig. 4. The effect of arterial lactate concentration on the change in arterio-venous concentration difference for glucose across the fetal brain. Points are plotted for all values obtained during the 12 lactate studies. Change in arterio-venous difference for glucose is calculated by subtracting the arterio-venous difference measured before beginning the infusion (0 h) in each animal from the values measured at 1, 2, 3 and 4 h.

pyruvate from the brain has been described as blood lactate levels begin to fall during recovery from anoxia in fetal and neonatal rats (Vannucci & Duffy, 1976).

Inhibition of brain glucose uptake during β -hydroxybutyrate infusions may occur by the same or a different mechanism. The initial fall in the arterio-venous difference for glucose parallels the rise in butyrate levels, and ketones have been described to inhibit brain glucose utilization *in vitro* in the neonatal rat brain (Fernandez & Medina, 1986). This apparently occurs by decreasing pyruvate oxidation (Itoh & Quastel, 1970) Booth & Clark, 1981; Miller, Kiney, Corrdry & Staton, 1982). Subsequently, the arterio-venous difference for glucose remains low (although not significantly different from the value at 0 h) as butyrate concentrations fall and lactate concentrations continue to rise (Fig. 2). This apparently incomplete inhibition of brain glucose uptake despite high lactate concentrations may partly be explained by the fact that the lactate levels only reached the lower range of the threshold concentrations which were associated with the fall in glucose uptake during lactate infusions. Furthermore, lactate and β -hydroxybutyrate compete for entry into the brain for a single monocarboxylic acid carrier (Conn *et al.*, 1983), so that brain entry of lactate would be expected to be lower and have less of an effect in the presence of β -hydroxybutyrate. The small efflux of pyruvate from the brain during the butyrate infusion, although not statistically significant, is consistent with both these mechanisms of action (Fig. 2).

The rise in arterial lactate concentrations observed during β -hydroxybutyrate infusion has been described previously (Miodovnik, Lavin, Harrington, Leung, Seeds & Clark, 1982 a & b; Paton, Levitsky & Fisher, 1989). However the origin of this lactate is not known. It does not appear to arise from the fetal

brain, nor from the fetal muscle (Harding & Charlton, unpublished observations). Preliminary observations suggest that the placenta may be a site of increased lactate production during butyrate infusions (Harding & Charlton, unpublished observations). Alternatively, inhibition of pyruvate decarboxylation in tissues other than brain, such as the fetal liver, may result in reduced fetal lactate consumption and therefore lactate accumulation.

In summary, these studies have failed to demonstrate significant fetal brain uptake of alternative substrates during short-term substrate infusions. However infusion of both lactate and β -hydroxybutyrate appears to inhibit fetal brain glucose uptake. It is interesting to speculate on the possible clinical significance of these observations. Inhibition of glucose metabolism suggests a possible mechanism whereby brain lactate concentrations exert their damaging effect during ischaemic injury (Myers, Wagner & de Courten-Myers, 1983; Rehnckrona, Rosen & Siesjo, 1981). Similarly, inhibition of brain glucose uptake by high circulating β -hydroxybutyrate concentrations suggests a possible mechanism for the observed poor neurological outcome of infants whose mothers suffered episodes of ketosis during pregnancy (Churchill, Berendes & Nemore, 1969).

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